

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i)		Substitution: work done = 450×0.8 (1) $\times 2$ (1) $= 720$ (1) [J] Answer 360 [J] award 2 marks Alternative: $2 \times 0.8 = 1.6$ [m] (1) 450×1.6 (1) $= 720$ (1) [J]	1 1	1		3	3	
		(ii)		Answer from (i) ecf expect 720 [J] or 720 [J]	1			1		
		(iii)		[Work done in overcoming] friction / lifting the forks or [energy transferred as] sound / heat		1		1		
	(b)			$450 - (12 \times 10) = 330$ [N] (1) $\frac{330}{10} = 33$ (1) [kg] Alternative: $\frac{450}{10} = 45$ (1) [kg] $45 - 12 = 33$ (1) [kg]		2		2	2	
				Question 1 total	3	4	0	7	5	0